

Quantum Gravity: 0/0 of Final Unification

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The Removable Singularity Framework

Abstract

We show that quantum gravity is the 0/0 of final unification: the resolution of every singularity. The Planck scale ($l_P = 1.6e-35$ m) is the 0/0 of physics, where quantum mechanics and general relativity both fail. Loop Quantum Gravity makes space discrete (area is quantized: $A = 8\pi l_P^2 \sqrt{j(j+1)}$) and resolves the Big Bang singularity via the Big Bounce. String theory shows gravity is EMERGENT from closed strings. AdS/CFT shows quantum gravity IS holographic. This connects the Big Bang, black holes, holography, measurement, and RMT.

1. The Incompatibility

Quantum mechanics describes small scales (atoms, particles). General relativity describes large scales (stars, galaxies). They are INCOMPATIBLE at the Planck scale -- this is the 0/0 of physics, where both theories fail.

2. Planck Scale

$$l_P = \sqrt{\hbar * G / c^3} = 1.616e-35 \text{ m}$$

The Planck scale is where quantum gravity dominates:

- Planck length: $1.616e-35$ meters
- Planck time: $5.391e-44$ seconds
- Planck mass: $2.176e-8$ kg (22 micrograms)

Below this scale, both QM and GR fail (0/0).

3. Loop Quantum Gravity

$$A = 8 * \pi * l_P^2 * \sqrt{j(j+1)}$$

Space is composed of QUANTA. Area is quantized -- it cannot be smaller than $\sim 10^{-70} \text{ m}^2$. The Big Bang singularity is RESOLVED: instead of a singularity, density reaches a maximum and BOUNCES (Big Bounce, $\rho_{\text{max}} = 0.41\rho_{\text{Planck}}$).

4. String Theory

Fundamental objects are 1-dimensional STRINGS, not points. Extra dimensions: 10-11 (not 4). Gravity EMERGES from closed strings (the graviton has spin-2). Gravity is NOT fundamental -- it is emergent.

5. AdS/CFT

$$D\text{-dim quantum gravity} = (D-1)\text{-dim CFT}$$

Quantum gravity in the D-dimensional bulk is equivalent to a (D-1)-dimensional conformal field theory on the boundary. This IS quantum gravity (holographic). Spacetime entropy $S = A/(4l_P^2)$: the universe is a quantum computer.

6. Connections

System	Prior Chapter	Connection
Big Bang	Ch.50 Big Bang	Singularity resolved
Black holes	Ch.32 Black holes	Horizon, info paradox
Holographic	Ch.47 Holographic	AdS/CFT
Measurement	Ch.49 Measurement	Quantum geometry
RMT	Ch.44 RMT	SYK model
Arrow of time	Ch.48 Arrow of time	Quantum time

7. Conclusion

Quantum gravity is the FINAL UNIFICATION: the 0/0 that resolves every singularity. The Planck scale is the 0/0 of physics. Loop Quantum Gravity quantizes space and replaces the Big Bang singularity with a bounce. String theory shows gravity is emergent. AdS/CFT shows quantum gravity is holographic. Space is information: the universe is a quantum computer. This connects the Big Bang, black holes, holography, measurement, and RMT -- resolving ALL 0/0 singularities.

References

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